Beyond BB84: The Cost of Trustless Detectors in Quantum Key Distribution

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Aug 2026

Abstract

Quantum Key Distribution (QKD) was first put forward in 1984 by Bennett and Brassard through their foundational proposal for bipartite secure key exchange taking advantage of quantum mechanics, now commonly known as BB84. In the following decades, a number of inspired methods have emerged to guarantee shared secrecy. In particular, Measurement-Device-Independent (MDI) QKD, presented in 2012, improves BB84 security assumptions by eliminating the need to trust photon detectors. In order to be able to deploy QKD at scale, providers must map the trade-offs of either protocol, and comparisons of the two in secure information exchange rate (key rate), cost and scalability allow for efficient application of this maturing technology. In this report, we aim to contribute to the solution of this problem through use of NetSquid, a quantum network discrete event simulation (DES) engine, and an algebraic cost model, to simulate and compare network behaviour for both MDI-QKD and BB84. We see that the price to pay for MDI's improved security is a drop in key rate, of up to 10 times, in modest metropolitan networks under realistic hardware assumptions.

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1 Introduction

The advent of large-scale quantum computing poses a fundamental threat to classical public-key cryptography. Modern key-exchange standards, including RSA [1] and Elliptic Curve Cryptography [2, 3], derive their security from the computational hardness of integer factorisation and the discrete logarithm problem. Shor's algorithm [4], published in 1997, demonstrated that both problems can be solved in polynomial time on a sufficiently powerful quantum computer, rendering these standards vulnerable to a quantum-capable adversary. Post-Quantum Cryptography (PQC) responds to this threat by identifying classically hard problems not susceptible to known quantum speed-ups; Quantum Key Distribution (QKD) takes the complementary route, using quantum mechanics to guarantee shared secrecy rather than computational assumption.

QKD was first conceived by Wiesner [5] in the 1970s and given its first operational form by Bennett and Brassard in 1984 [6]. In the decades since, QKD has matured into one of the most field-tested quantum technologies, with national-scale fibre deployments serving over 150 users across thousands of kilometres [7] and metropolitan testbeds demonstrating sustained multi-vendor operation [8, 9]. This trajectory has motivated a growing body of research into how QKD protocols should be chosen and deployed at scale. This work focuses on two protocols that represent a natural axis of comparison: BB84, the original Prepare-and-Measure scheme, and Measurement-Device-Independent QKD (MDI-QKD), proposed by Lo, Curty and Qi in 2012 [10]. Their protocols and theoretical performance bounds are introduced in Section 2.

MDI-QKD improves upon BB84's security model by removing the need to trust photon detectors, an assumption that is difficult to verify in practice and has been exploited in hardware-level attacks. This advantage, however, comes at a real cost in key rate: at metropolitan distances, BB84 consistently delivers rates an order of magnitude higher than MDI under comparable hardware conditions [11, 12]. For network providers selecting between the two protocols, this trade-off must be mapped carefully across distance, topology, and network scale. The question is not simply which protocol is more secure, but under what deployment conditions MDI's security advantage justifies its rate penalty, and at what economic and energetic cost.

Despite the commercial momentum behind both protocols — evidenced by field deployments [13, 14], heterogeneous network testbeds [15], and European infrastructure programmes [16] — the quantitative basis for choosing between BB84 and MDI-QKD for a given application remains poorly characterised. Simulation of MDI-QKD at metropolitan scale has not systematically explored the interplay between topology, relay placement, and hardware parameters across a sufficiently wide parameter space to support confident protocol selection [17]. Cost models for MDI networks are largely absent from the literature: whilst integer linear programming approaches have been developed for PM and entanglement-based QKD infrastructure [18], their extension to MDI has not been carried out. Energetic footprint, an increasingly central criterion in telecommunications infrastructure procurement, has only recently begun to receive attention in the QKD context [19]. A rigorous parametric comparison across key rate, cost, and scalability is absent.

This work addresses these gaps through a two-part methodology. First, we implement and simulate both protocols within NetSquid [20], a discrete-event quantum network simulation engine, across a ten-layer hardware model spanning ideal to physically realistic conditions. Second, we model deployment costs algebraically from per-component hardware pricing for both protocols across a range of network sizes. We compare BB84 and MDI-QKD across three dimensions: key rate as a function of distance and hardware parameters; deployment cost under realistic component costs; and scalability, defined as the marginal key rate and cost when adding users to a fixed metropolitan topology. The remainder of this report is structured as follows. Section 2 introduces the protocols, their hardware underpinnings, and the theoretical performance bounds that govern them. Section 3 describes the simulation architecture and cost model. Section 4 presents results across all three comparison axes, and Section 6 concludes with deployment recommendations and directions for future work.

2 Shared Quantum Secrecy

Quantum Key Distribution refers to any communications protocol in which two or more parties use quantum channels, the sharing of quantum states, together with authenticated classical channels and post-processing, to establish a shared secret cryptographic key. The security of such a scheme is guaranteed by the laws of quantum mechanics, providing information-theoretic security against any eavesdropper whose capabilities are bounded by quantum physics. Since Bennett and Brassard's foundational paper in 1984 [6], numerous variants of QKD have been proposed. This work focuses on two: the original Prepare-and-Measure scheme (BB84) and Measurement-Device-Independent QKD (MDI-QKD), proposed by Lo, Curty and Qi in 2012 [10].

2.1 BB84

BB84 is representative of the class of Prepare-and-Measure (PM) protocols. Alice begins by preparing a string of n photons, each encoded in one of four polarisation states ($0°$, $45°$, $90°$ or $135°$) selected uniformly at random. In the qubit formalism, these correspond to $|0\rangle$, $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|1\rangle$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, respectively. Alice selects a basis ($\{|0\rangle, |1\rangle\}$ or $\{|+\rangle, |-\rangle\}$) for each qubit uniformly and independently, then uniformly and independently again, she applies a bit flip within that basis before transmission.

She transmits these quantum states to Bob over a quantum channel. Bob measures each photon in a uniformly and independently random basis. Alice and Bob then use an authenticated classical channel to compare basis choices and discard all photons measured in mismatching bases, producing a raw, sifted key. A portion of this sifted key is compared over the classical channel to estimate the Quantum Bit Error Rate (QBER): the fraction of photon pairs measured in the same basis that nevertheless disagree. A QBER below 11% is taken as evidence that the channel has not been compromised [21]. Alice and Bob then apply error correction and privacy amplification (hash-based post-processing that shortens the raw key, removing any partial information an eavesdropper may have gathered) to produce the final shared secret key. Figure 1 illustrates the basic bipartite logical layer.

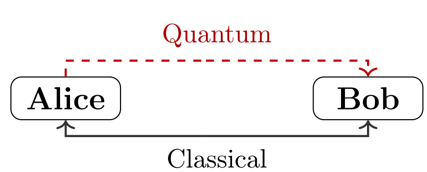

Figure 1: Communication diagram for point-to-point BB84. Alice sends quantum states to Bob; basis reconciliation and post-processing occur over authenticated classical channels.

Security proofs for BB84, and QKD generally, are conditional on the modelling assumptions made about devices, network, and attacker capabilities. A particular vulnerability in practice arises from weak coherent pulse (WCP) laser sources, whose photon number per pulse follows a Poisson distribution. Multi-photon pulses allow a Photon Number Splitting (PNS) attack [22], in which Eve intercepts and holds trailing photons until bases are announced, extracting key information without disturbing the QBER. The decoy-state technique [23] mitigates this by varying pulse intensity randomly, enabling Alice and Bob to tightly bound the single-photon contribution to their key.

2.2 Measurement-Device-Independent QKD

MDI-QKD addresses a fundamental vulnerability shared by all PM schemes: the need to trust the measurement devices at Bob's node. Side-channel attacks on detectors represent a significant practical threat, as the gap between idealised device models assumed in security proofs and real hardware is difficult to close. MDI removes detectors entirely from the trust boundary by introducing an untrusted third-party relay, Charlie, who performs all measurement operations.

In MDI-QKD, both Alice and Bob independently prepare photon strings in the manner of BB84 and transmit these to Charlie. Charlie performs a Bell State Measurement (BSM) on each arriving photon pair, projecting onto the two-qubit Bell basis:

$$|\phi^{\pm}\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \qquad |\psi^{\pm}\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}.$$

Charlie announces the measurement outcome over a classical channel to both Alice and Bob. Since the BSM measures only the *joint correlation* between the two incoming photons rather than their individual polarisations, an eavesdropper with full control of Charlie's apparatus can learn nothing about Alice's or Bob's individual bit values. The physical mechanism underlying the BSM is Hong-Ou-Mandel (HOM) interference [24]: two indistinguishable photons arriving simultaneously at a 50:50 beam splitter always exit the same output port due to bosonic quantum interference. Coincident clicks at the downstream polarising beam splitters then identify the projected Bell state. Photon indistinguishability is therefore a prerequisite for MDI-QKD: timing jitter, spectral mismatch, and polarisation misalignment all reduce HOM visibility and degrade BSM success probability. The bipartite logical layer for MDI-QKD is shown in Figure 2.

Figure 2: Communication diagram for point-to-point MDI-QKD. Both Alice and Bob transmit quantum states to Charlie; Charlie performs a BSM and announces the result classically to both parties.

Alice and Bob sift by matching bases as in BB84. Upon receiving the BSM outcome, each party knows their own basis choice and bit value, and can deduce the other's bit via the outcome. Depending on the Bell state announced, one party selectively bit-flips to ensure agreement, as summarised in Table 1. In the linear-optical BSM implementation, only two of the four Bell states ($|\psi^+\rangle$ and $|\psi^-\rangle$) are unambiguously distinguishable [10]; detection events corresponding to $|\phi^\pm\rangle$ are discarded as inconclusive. Privacy amplification and error correction then proceed as in BB84.

BSM outcome	$\|\psi^+\rangle$	$\|\psi^-\rangle$
Diagonal basis	No bit flip	Bit flip
Rectilinear basis	Bit flip	Bit flip

Table 1: Bit-flip rules upon receipt of BSM outcome [10]. Alice and Bob agree in advance which party applies the flip; if both flip, there is no net change.

The abstract operation of the BSM in quantum information terms is shown in Figure 3: a CNOT gate followed by a Hadamard and measurement. The physical implementation uses a 50:50 beam-splitter and polarising beam-splitters, with the Hong-Ou-Mandel effect ensuring that only coincidence detection events, two simultaneous clicks at the detector array, constitute valid BSM outcomes.

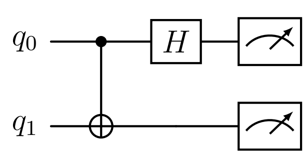

Figure 3: Bell State Measurement circuit: CNOT gate followed by Hadamard, then measurement in the computational basis.

MDI-QKD's security advantage comes at a cost. Key generation requires a successful coincidence detection at Charlie – both photons from Alice and Bob must arrive and be detected within the same time window. This requirement, combined with independent connection losses on each arm, imposes a rate penalty relative to BB84's single-link loss

[10]. MDI's QBER tolerance is also more sensitive than BB84's, as two independent quantum channels each contribute noise, enlarging the attack surface accounted for by security proofs.

2.3 Theoretical Performance Bounds

The maximum achievable secure key rate in any point-to-point (no quantum repeater) QKD scheme is bounded by the Pirandola–Laurenza–Ottaviani–Banchi (PLOB) bound [25]. This fundamental result establishes that, against all attacks permitted by quantum physics, the key rate of a repeater-less link scales at most linearly with channel transmittance η:

$$R \leq -\log_2(1-\eta) \approx \eta \log_2 e \quad \text{for } \eta \ll 1.$$

Since transmittance decays exponentially with fibre length L as $\eta = 10^{-\alpha L/10}$ (with α the fibre attenuation coefficient in dB/km), the PLOB bound implies that key rates must fall exponentially with distance. Both BB84 and standard MDI-QKD are subject to this bound.

Twin-Field QKD (TF-QKD), proposed in 2018 by Lucamarini et al. [26], exploits the same untrusted-relay tenet as MDI but replaces the two-photon BSM with single-photon interference. The result is that key rates scale as $\mathcal{O}(\sqrt{\eta})$ rather than $\mathcal{O}(\eta)$, exceeding the PLOB bound and extending the distance at which secure communication is achievable. TF-QKD thus represents the most practically significant near-term stepping stone beyond the repeater-less limit, and motivates the question central to this work, of precisely where and under what conditions MDI-QKD's security advantage over BB84 justifies its rate penalty, before that regime is superseded by TF-QKD or, eventually, quantum repeaters.

3 Methodology

We compare BB84 and MDI-QKD across three dimensions: performance, deployment cost, and scalability. Together these axes capture the trade-off a network provider faces when selecting between the two protocols for a given metropolitan deployment.

Performance. The primary metric is secure key rate, measured in bits per second: the length of the final distilled key divided by the time taken to produce it. Since all QKD-generated keys are information-theoretically equivalent in security regardless of protocol, key rate is the sole performance discriminant between BB84 and MDI-QKD. We simulate key rate as a function of distance and individual hardware parameters using NetSquid [20], a discrete-event quantum network simulation engine, across a ten-layer hardware model that progressively introduces realistic physical imperfections from an ideal baseline.

Cost. Deployment cost is modelled algebraically from per-component hardware costs (photon sources, single-photon detectors, beam splitters, and fibre), with the number of each component determined by the network topology under each protocol. For a given user count and area, total infrastructure cost follows directly from the component counts implied by each protocol's topology. Beyond simple cost curves, we frame deployment decisions as a cost–rate optimisation: a provider operating under a fixed capital budget seeks the configuration maximising average key rate, whilst a provider with a minimum key rate requirement seeks the cheapest configuration meeting it. We sweep relay count K and compute the (cost, average key rate) pair per configuration, identifying deployment-optimal choices for each protocol.

Scalability. We define scalability as the marginal key rate and cost incurred when adding a user to an existing network. Since BB84 requires a direct link between every user pair whilst MDI routes traffic through relay nodes, the two protocols exhibit fundamentally different scaling behaviour. We quantify this by sweeping user count at fixed topology parameters and extracting per-user marginal rates and costs for both protocols.

3.1 Cost Model

We model the total deployment cost of a QKD network algebraically from per-component hardware costs. All costs are capital expenditure (CapEx) only; operational expenditure — in particular the cryogenic cooling required by superconducting nanowire single-photon detectors (SNSPDs) and quantum dot (QD) sources — is not modelled. Classical communication infrastructure (basis reconciliation, error correction, and key forwarding) is assumed to be a sunk cost of existing network infrastructure. The most directly related prior cost analysis is that of Karavias *et al.* [18], which applies integer linear programming to PM-QKD and entanglement-based infrastructure planning; MDI-QKD is not considered there, and cost–rate trade-offs are not explored. We extend this cost-modelling perspective to MDI-QKD and a third protocol, trusted-node BB84 (TBB84), described below.

3.1.1 Component Counts

The two network topologies impose fundamentally different hardware structures. In BB84, every user pair requires a dedicated direct dark-fibre link: the network is a complete mesh of $N(N-1)/2$ quantum channels. At each receiver, a polarising beam splitter (PBS) separates Z and X basis outputs to two single-photon detectors (SPDs), and an electro-optic modulator (EOM) performs active basis selection. No relay infrastructure is required.

In MDI-QKD, users transmit photons to BSM relay stations rather than to each other. Each relay is a passive station comprising a 50:50 HOM beam splitter, two PBS (one per arm), four SPDs, and one optical switch for routing photons from cross-cluster pairs. The network requires N user-to-relay fibre links and $K(K-1)/2$ relay-to-relay backbone links.

Table 2 summarises the component counts for both protocols.

Component	BB84	MDI
Single-photon sources	N	N
Single-photon detectors	$2N$	$4K$
50:50 beam splitter (HOM)	0	K
Polarising beam splitter	N	$2K$
Electro-optic modulator	N	0
Optical switch	0	K
Fibre links	$\dfrac{N(N-1)}{2}$	$N+\dfrac{K(K-1)}{2}$

Table 2: Hardware component counts for a network of N users and K relay nodes. MDI relay nodes are passive BSM stations. BB84 forms a direct mesh with no relay infrastructure.

Fibre cost scales as $\mathcal{O}(N^2)$ for BB84 and as $\mathcal{O}(N)$ for MDI at fixed K, providing the structural cost incentive for relay-based architectures as user counts grow.

3.1.2 Single-Photon Detector Cost

SPD cost depends primarily on detection efficiency and technology class. We model it as a linear function of detection efficiency η, anchored at two IDQ product operating points:

the ID230 InGaAs SPAD ($\eta = 0.20$, dark count 200 cps at £15,000 placeholder) and the ID281 SNSPD ($\eta = 0.90$, dark count 50 cps at £100,000 placeholder):

$$C_{\mathrm{SPD}}(\eta) = \frac{100{,}000 - 15{,}000}{0.90 - 0.20}(\eta - 0.20) + 15{,}000. \tag{1}$$

Both anchor prices are placeholder estimates; a quote from ID Quantique has been requested. Note that for the ID230 SPAD, detection efficiency and dark count rate are not independent: operating the detector at higher bias increases both efficiency and dark count simultaneously (e.g., $\eta = 0.10$ yields $\sim$80 cps whilst $\eta = 0.20$ yields $\sim$200 cps). The simulation treats η and d_c as independent parameters to isolate the marginal effect of each, but the physically consistent SPAD operating point pairs $\eta = 0.20$ with $d_c = 200$ cps. This linear model is a simplification: SPD cost in practice is not strictly linear in η, and dark count rate is a second major performance parameter that is not independently captured. The model is adequate for comparative analysis when sweeping between SPAD and SNSPD technology classes.

A further consideration is that commercial SNSPD products are typically sold as multi-channel units integrating several detector channels within a single cryostat (for example, the ID Quantique product range offers 2–8 channels per unit). An MDI relay station requiring four SPDs (two polarising beam splitter outputs per arm) may therefore be equipped with a single multi-channel unit rather than four independent detectors. In this case the cost per relay is one unit price rather than $4 \times C_{\mathrm{SPD}}(\eta)$; the per-channel model overestimates relay detector cost and will be revised on confirmation of unit pricing.

Default component costs are listed in Table 3; all values are estimates pending datasheet verification.

Component	Default cost	Note
QD photon source	£150,000	Pending verification
SPD (IDQ ID230 SPAD, $\eta = 0.20$, $d_c \approx 200$ cps)	£15,000	Placeholder; IDQ quote requested
SPD (IDQ ID281 SNSPD, $\eta = 0.90$, $d_c \approx 50$ cps)	£100,000	Placeholder; IDQ quote requested
50:50 HOM beam splitter	£1,000	Pending verification
Polarising beam splitter	£500	Pending verification
Electro-optic modulator	£2,000	Pending verification
Optical switch	£5,000	Pending verification
Dark fibre (installed)	£10,000/km	Region-dependent; pending verificat

Table 3: Default hardware component costs used in the cost model. All values are unverified estimates.

3.1.3 Total Deployment Cost and Fibre Tortuosity

The total deployment cost is:

$$C_{\text{total}} = C_{\text{hardware}} + C_{\text{fibre}}, \tag{2}$$

where

$$C_{\text{hardware}} = N_s C_s + N_d\, C_{\text{SPD}}(\eta) + N_{\text{BS}} C_{\text{BS}} + N_{\text{PBS}} C_{\text{PBS}} + N_{\text{EOM}} C_{\text{EOM}} + N_{\text{sw}} C_{\text{sw}}, \tag{3}$$

and $C_{\text{fibre}} = L_{\text{total}} \cdot C_f$, with L_{total} the total fibre deployed in km and $C_f = 10{,}000/\text{km}$.

In practice, installed cable follows duct routes longer than the straight-line distance between nodes. We model this through per-link tortuosity: each physical cable (i, j) is assigned an independent factor τ_{ij} drawn from a truncated normal distribution:

$$\tau_{ij} \sim \max\big(1,\ \mathcal{N}(\mu_\tau,\, \sigma_\tau^2)\big)\,, \qquad \mu_\tau = 1.2, \quad \sigma_\tau = 0.1. \tag{4}$$

The mean $\mu_\tau = 1.2$ is representative of urban cable routing, where installed fibre is typically 15–25% longer than the Euclidean distance. Each physical cable draws one factor, held constant across all simulation runs. Tortuosity affects both the simulated key rate (via increased fibre loss on longer cables) and the cost model (via increased fibre deployed). For MDI, user-to-relay and relay-to-relay cables each draw a single factor, shared across all user pairs that traverse that cable; this ensures physical consistency when the same cable appears in multiple end-to-end paths.

3.1.4 Cost–Rate Optimisation Framework

Beyond simple cost curves, we frame deployment decisions as a cost–rate optimisation. For a provider with capital budget $C_{\max}$, the optimal configuration maximises average key rate $\bar{R}$ subject to $C_{\text{total}} \leq C_{\max}$. For a provider with minimum rate requirement $R_{\min}$, the optimal configuration minimises C_{total} subject to $\bar{R} \geq R_{\min}$. By sweeping relay count K at fixed N and area under each protocol, we obtain a (cost, key rate) frontier per protocol. Both optimisation problems can be read directly from this frontier, enabling protocol-agnostic deployment recommendations.

3.2 Performance Analysis

The primary performance metric is the secure key rate R, measured in bits per second: the key length extracted by a simulation run divided by the simulation duration. Since all QKD-generated keys are information-theoretically equivalent in security regardless of protocol, key rate is the sole performance discriminant between BB84 and MDI-QKD for a given network. A symmetric QBER threshold of 11% is applied: simulation runs with QBER above this value are discarded as failures [21]. All key rates are computed in the asymptotic regime, without finite-key corrections; this overestimates extractable key at low photon count rates but is standard in comparative simulation studies at metropolitan scale.

3.2.1 Hardware Model

Physical impairments are introduced via a ten-layer parametric model. Each layer adds one impairment cumulatively to the previous, from a fully ideal baseline (layer 0) to a system incorporating all realistic imperfections (layer 9). This structure allows the marginal impact of each physical parameter on key rate to be isolated, forming the basis of the parameter sweep analysis in Section 4. Table 4 lists the ten layers.

Layer	Parameter added	Symbol	Default value	Verification source
0	Ideal baseline	—	All perfect	—
1	Fibre attenuation	α	0.18 dB/km	Corning SMF-28 datasheet
2	Detector efficiency (Z basis)	η_Z	0.90	IDQ ID281 SNSPD datasheet
3	Dark count rate	d_c	50 cps	IDQ ID281 SNSPD datasheet
4	TX insertion loss	L_i	0.10 (10%)	
5	RX node loss	Δ_n	BB84: 2.0 dB; MDI: 1.0 dB/arm	
6	Source error rate	ε_s	0.015	Quandela Prometheus datasheet ($g^{(2)}(0) < 0.03$)
7	Fibre dephasing	β	10^{-4} ns^{-1}	
8	X-basis detector bias	η_X	0.85	
9	BSM beam-splitter efficiency	η_{BS}	0.97 (MDI only)	

Table 4: Ten-layer hardware model. Layer k includes all impairments of layer $k-1$ plus the new parameter listed. Layer 9 is the realistic operating point used for network simulations. Empty cells in the verification column indicate parameters pending datasheet confirmation.

Layer 5 introduces protocol-specific receiver node loss. For BB84, the RX path passes through an EOM (≈1.5 dB) and a PBS (≈0.5 dB), giving $\Delta_n^{\mathrm{BB84}} = 2.0$ dB. For MDI, each arm encounters only a PBS and HOM coupler excess loss without an EOM, giving $\Delta_n^{\mathrm{MDI}} = 1.0$ dB per arm; the two arms contribute independently to photon loss. Both values are indicative midpoints pending datasheet verification. Layer 9 (beam-splitter efficiency) applies to MDI only, where a non-ideal HOM split ratio ($\eta_{\mathrm{BS}} = 0.97$) reduces coincidence rate. Ideal single-photon sources are used throughout; source imperfection is captured solely by ε_s at layer 6. Weak coherent pulse (WCP) sources and decoy-state techniques [23] are not modelled, so reported rates represent an upper bound relative to practical WCP deployments.

3.2.2 NetSquid Simulation Engine

NetSquid [20] is a discrete-event simulation (DES) engine for quantum networks developed at QuTech. In a DES framework, system state changes only at discrete event times; between events no computation is performed. This suits photon-by-photon QKD simulation, where the relevant events are emission, propagation delay, detection, and classical post-processing. NetSquid represents quantum states as density matrices acted on by quantum channels, enabling physically accurate treatment of loss, noise, and measurement.

We drive NetSquid through a multiprocessing pool. Each worker runs one independent QKD simulation trial, returning key length, QBER, and wall-clock time. The main process distributes trials across worker processes using Python's `multiprocessing` module with spawn semantics, avoiding shared-state contamination from NetSquid's process-level global state. The pool is sized to 80% of available CPU cores by default. Results are collected in the main process and aggregated over trials. All results are persisted to an SQLite database (`results.db`) via a dedicated library (`lib/db.py`). Two tables are used: `p2p_results` stores one row per simulation trial (key rate, QBER, all physical parameters); `network_results` stores one aggregated row per network configuration (mean key rate, success rate, deployment cost, topology parameters). This separation enables post-hoc analysis without re-simulation.

3.2.3 Bipartite Single-Link Analysis

The bipartite simulation sweeps one hardware parameter at a time, holding all others at their layer-5 defaults. For each parameter value, $r = 20$ independent Monte Carlo trials are run; each trial constitutes one complete QKD exchange from photon emission to key extraction. The mean key rate and success rate (fraction of trials with QBER $< 11\%$) are reported as a function of the swept parameter.

In the BB84 simulation, Alice prepares $n = 1024$ photons per trial at $f = 10^7$ Hz. Each photon is assigned an independent random basis and bit value; Bob measures each received photon in a randomly selected basis. Channel loss, dark counts, and measurement noise are modelled as stochastic events within the DES engine. After each trial the key length (matching-basis, post-error-corrected bits) and simulation time are recorded; key rate is their ratio.

In the MDI simulation, Alice and Bob independently prepare photon strings and transmit to Charlie's relay. The simulation models the additional loss on both arms, the HOM visibility (governed by η_{BS}), and the BSM success probability. The linear-optical BSM discriminates $|\psi^+\rangle$ and $|\psi^-\rangle$ but cannot distinguish $|\phi^\pm\rangle$ [10]; approximately half of all photon pair arrivals yield inconclusive results and are discarded. The Charlie position along the link is a free parameter; in the network setting it is fixed by the relay's geometric position relative to the pair's users.

3.2.4 Multi-User Network Analysis

The multi-user simulation models a metropolitan network of N users within an $A \times A$ km grid. Users are placed uniformly at random. Relay positions are determined by k-means clustering of user positions, placing each relay at the centroid of its cluster. The centroid minimises the sum of squared user-to-relay distances within each cluster, directly reducing the mean user-to-relay path loss. For $K = 1$ the centroid equals the single-relay optimum; k-means generalises this to $K > 1$.

To decouple topology variability from Monte Carlo noise, all network experiments are repeated across multiple independently drawn user placements (seeds). Relay positions are re-optimised per seed. Aggregate statistics across seeds produce error bars that characterise topology uncertainty rather than sampling noise.

This work focuses on metropolitan-scale networks, where inter-node distances are short enough for both BB84 and MDI-QKD to establish keys at viable rates without trusted forwarding. Trusted-node relay architectures (such as trusted-node BB84) are therefore out of scope; their value arises at wide-area scales where direct and MDI links become loss-limited, a regime not addressed here. Two network protocols are simulated:

BB84 mesh. Every user pair (i, j) communicates over a dedicated direct link of tortuous length $d_{ij}\,\tau_{ij}$. All $N(N-1)/2$ pairs are simulated in parallel; r Monte Carlo trials are run per pair. Network key rate is the mean over all valid (pair, trial) combinations.

MDI network. Each user connects to their nearest relay. A pair (i, j) where both users share the same relay station runs an MDI simulation with Alice-to-relay distance $d_{iR}\,\tau_{iR}$ and Bob-to-relay distance $d_{jR}\,\tau_{jR}$. A cross-cluster pair additionally routes through the relay-to-relay backbone: Bob's effective path is $d_{jR}\,\tau_{jR}$ (user to serving relay R_j) plus $d_{R_jR_i}\,\tau_{R_jR_i}$ (backbone leg to Alice's relay R_i), and the serving relay applies an optical switch (insertion loss $\Delta_{\mathrm{sw}} = 1.0$ dB). The Charlie position is set to $d_{iR}/(d_{iR} + \text{total Bob path})$.

3.3 Scalability Analysis

Scalability is characterised through two orthogonal sweeps.

User count sweep. Relay positions are optimised for the maximum N in the sweep and held fixed as N varies from 2 to N_{max}. For each N, both protocols are simulated and their mean key rate and total deployment cost are recorded. The marginal rate $\Delta\bar{R}/\Delta N$

and marginal cost $\Delta C/\Delta N$ are computed numerically from adjacent sweep points. MDI amortises relay infrastructure cost as N grows, since relay hardware is fixed whilst user count increases; BB84 incurs one new fibre link per user added, giving $\mathcal{O}(N)$ marginal fibre cost growth.

Relay count sweep. User count N is fixed and K is varied from 1 to $K_{\max}$. BB84 is relay-independent and appears as a constant baseline. MDI key rates improve as K increases (shorter mean user-to-relay distances) until the relay-to-relay backbone becomes the dominant bottleneck; cost increases monotonically with K due to relay hardware. This sweep surfaces the deployment-optimal relay count for a given N and area, and provides the (cost, key rate) frontier used for cost–rate optimisation.

4 Results

bunch of graphs...

4.1 Deployment Recommendations

Cost–rate optimisation provides a direct basis for protocol selection. For each protocol, we sweep relay count K at fixed user count and area, computing the (total deployment cost, average key rate) pair for each configuration.

Two deployment questions are answered from this sweep. Given a capital budget $C_{\max}$, the optimal configuration is the one with the highest key rate subject to cost $\leq C_{\max}$. Given a minimum key rate requirement $R_{\min}$, the optimal configuration is the lowest-cost configuration meeting rate $\geq R_{\min}$. Comparing BB84 and MDI-QKD directly across this sweep reveals the regimes in which MDI's security advantage can be obtained within the same budget envelope as BB84, and the configurations for which MDI's rate penalty is acceptable given a stated security requirement.

5 Discussion

The results presented in Section 4 support a consistent picture of the trade-off space between BB84 and MDI-QKD. We discuss the key findings, their physical interpretation, the limitations of the methodology, and their implications for protocol selection.

5.1 The Security–Rate Trade-off

MDI-QKD's elimination of detector side-channel attacks comes at a key rate penalty rooted in its physical architecture. Whereas BB84 requires one photon to traverse the full link and be detected, MDI requires two photons — one from each party — to arrive at Charlie's relay simultaneously and produce a coincidence event. Loss on each arm compounds multiplicatively: if the single-arm transmittance is η_{arm}, the BSM detection probability scales as $\eta_{\text{arm}}^2\,\eta_{\text{BS}}\,\eta_d^2$, compared to $\eta_{\text{link}}\,\eta_d$ for BB84. At metropolitan distances and typical detector efficiencies, this effectively squares the channel transmittance, consistent with the order-of-magnitude rate gap between the two protocols in the bipartite setting.

At the network level, the rate gap between MDI and BB84 widens relative to the bipartite case. MDI users route photons to a relay rather than directly to their communication partner, incurring a geometric detour for cross-cluster pairs. This routing overhead is absent in BB84's direct mesh. The relay distance advantage is a function of relay count K and user distribution; the relay sweep quantifies the extent to which increasing K can recover the MDI rate penalty.

5.2 Cost Model Limitations

The cost model is a first-order CapEx-only estimate. Several factors are not captured.

Operational expenditure. Cryogenic systems — both SNSPD detectors and QD sources — require continuous cooling to temperatures of order 2–4 K, incurring energy costs that can rival hardware CapEx over a ten-year deployment lifetime [19]. The energetic cost of operating K SNSPD-equipped MDI relay stations may be significant at scale and is not captured by the CapEx-only model.

Cost–simulation decoupling. The cost model and the simulation model are not tightly coupled. Adding components to the cost model raises cost, but the simulated key rate responds only through changes in topology geometry. The insertion loss of individual components (EOM, PBS, optical switch) is captured in the protocol-specific node loss Δ_n, but this is a fixed per-protocol constant rather than a value derived from the component cost model at runtime. A cheaper optical switch with higher insertion loss would reduce key rate without reducing cost, a trade-off invisible to the cost model as presented.

BB84 fibre scaling at large N. The BB84 direct mesh assumes one independent dark-fibre run per user pair. In practice, WDM or switched architectures would serve many pairs over shared fibre infrastructure at high N, substantially reducing BB84 fibre cost. The $\mathcal{O}(N^2)$ fibre scaling should therefore be interpreted as an upper bound applicable to passive, unmanaged dark-fibre mesh deployments; it is physically exact for that specific case.

5.3 Simulation Model Limitations

Source brightness and insertion loss decoupling. The Quandela Prometheus single-photon source used as the reference product operates at 975 nm and achieves a minimum fibred brightness of 40%, with frequency conversion required to reach the 1550 nm telecom band. In the simulation, source brightness is absorbed into the effective source repetition rate (`source_freq` $= 10^7$ Hz, consistent with the 32 MHz raw rate after 40% brightness), and the `init_loss` parameter models separate optical path losses independently. Frequency conversion efficiency, additional conversion-stage noise, and the explicit coupling between source brightness and photon rate are not separately modelled. This is a deliberate simplification; a full model would propagate each loss stage independently and account for Raman background photons introduced by the conversion stage.

Ideal single-photon sources. All simulations assume one photon per pulse with no multi-photon component. WCP laser sources are overwhelmingly more practical and affordable than QD sources, and their multi-photon components require decoy-state analysis [23] to extract secure key rate. The ideal source assumption gives an upper bound on key rate and a lower bound on source cost; both should be re-evaluated for WCP deployments.

Asymptotic key rates. Finite-key corrections become significant at low photon count rates and short block lengths, reducing the extractable key compared to the asymptotic rate reported here. At long distances where photon detection events are sparse, the finite-key penalty is expected to be most severe for MDI-QKD, potentially widening the gap with BB84 beyond what the asymptotic analysis suggests.

Relay placement. k-means relay placement minimises mean user-to-relay distance but is not provably optimal for network key rate or cost. ILP-based placement [18] could provide tighter lower bounds on minimum MDI deployment cost, and would be a valuable extension of this work.

Metropolitan network scope. All simulations model networks within a 10×10 km metropolitan grid, where inter-node distances are short enough for both BB84 and MDI-QKD to establish keys at useful rates without range-extending mechanisms. At wider-area scales — city-to-city or national-backbone links — direct and MDI-QKD links become loss-limited and two distinct range-extending architectures become relevant. Trusted-node relay networks forward classical XOR-combined keys across intermediate nodes; this restores range at the cost of end-to-end security, since a compromised relay node exposes the key. Quantum repeaters, by contrast, extend range through entanglement swapping and distillation, maintaining end-to-end quantum security without trusted intermediaries. Quantum repeaters remain a longer-term technology requiring quantum memories and mid-circuit measurements at commercially unproven fidelity and scale. This work does not model either architecture; the metropolitan scope is a deliberate choice that allows direct protocol comparison without the additional variables introduced by relay forwarding or repeater chain design.

5.4 Implications for Protocol Selection

The results support a protocol-selection framework based on two axes: security requirement and scale. At small N (fewer than approximately [N*] users), BB84's direct mesh is both cheaper and faster than MDI. As N grows, BB84 fibre cost scales as $\mathcal{O}(N^2)$ whilst MDI scales as $\mathcal{O}(N)$; MDI becomes cost-competitive at [N*] users, a crossover that depends on area, relay count, and component prices. Where the deployment scale exceeds this crossover, the choice between BB84 and MDI reduces to a security question: MDI eliminates detector side-channel vulnerabilities at the cost of a lower key rate; BB84 delivers higher throughput within the trust assumptions of its detector model.

6 Conclusion

6.1 Summary of Contributions

This work has built a complete simulation and cost modelling infrastructure for the comparative analysis of BB84 and MDI-QKD at metropolitan scale. On the simulation side, both protocols are implemented within the NetSquid discrete-event simulation engine under a ten-layer hardware model that progressively introduces physical imperfections from an idealised baseline to a fully realistic system. Point-to-point key rates are obtained via Monte Carlo simulation across a parameter space covering fibre loss, detector efficiency, dark count rate, TX insertion loss, receiver node loss, source error rate, fibre dephasing, detector basis bias, and BSM beam-splitter efficiency. The simulation infrastructure parallelises across CPU cores using a spawn-context pool and persists all results to a structured SQLite database for post-hoc analysis without re-simulation.

At the network level, both protocols are implemented as multi-user simulators: a direct BB84 mesh and an MDI-QKD network with relay BSM stations. Relay positions are determined by k-means clustering of random user placements, and all network experiments are repeated across multiple independent seeds to characterise topology variability. A cost model computes total CapEx algebraically from per-component hardware costs and topology-determined component counts, incorporating a stochastic fibre tortuosity model that accounts for cable routing overhead in both the cost and the simulated key rate.

6.2 Findings

[This section will be completed once simulation results are generated. The following bullet points identify the expected finding structure.]

- The MDI-QKD key rate is approximately [X]$\times$ lower than BB84 in the bipartite setting at metropolitan distances under layer-9 hardware, consistent with the squaring of channel transmittance imposed by the BSM coincidence requirement.
- The rate gap increases in the network setting due to cross-cluster relay routing detours; the magnitude of the increase depends on relay count and user distribution.
- MDI total deployment cost is lower than BB84 for $N > N^*$ users due to the $\mathcal{O}(N)$ vs $\mathcal{O}(N^2)$ fibre scaling; the crossover N^* depends on relay hardware cost, area, and fibre cost per km.
- Cost–rate optimisation reveals a deployment-optimal relay count for MDI; the optimal K increases with N and area.

- Detector efficiency is the dominant hardware parameter limiting key rate in the bipartite setting at short and medium distances; fibre attenuation dominates at long distances.

6.3 Future Work

Several extensions are of immediate interest.

Finite-key corrections. All key rates are asymptotic; finite-key bounds [27] would reduce reported rates and alter the comparison, particularly at long distances where detection events are sparse and block lengths are short.

Weak coherent pulse sources. Replacing ideal QD sources with WCP laser sources modelled via the decoy-state technique [23] would give rates more directly comparable to published experimental deployments [11, 13, 14] and would substantially reduce assumed source cost.

Optimal relay placement. k-means relay placement is heuristic. ILP-based placement [18] solves the joint relay placement and routing problem to global optimality, providing tighter lower bounds on minimum MDI deployment cost.

Real network topologies. Extending the simulator to accept fixed network graphs from real city infrastructure would enable direct comparison with published deployed networks [11, 13, 14] and provide a practical planning tool for network operators.

Twin-Field QKD. TF-QKD and its variants [26] achieve key rates scaling as $\mathcal{O}(\sqrt{\eta})$, surpassing the PLOB bound and extending the distance at which MDI-level security is achievable without quantum repeaters. Extending this framework to TF-QKD would complete the repeater-less QKD comparison.

Operational expenditure. Incorporating cryogenic OpEx for SNSPD-equipped deployments, using the energetic analysis methodology of [19], would enable full lifetime cost comparisons and may substantially alter the economic case for MDI at high relay counts.

Trusted-node relay networks. At wide-area scales where metropolitan link distances are exceeded and direct MDI-QKD links become loss-limited, trusted-node relay architectures become relevant. Extending the cost model and simulation to trusted-node BB84 would enable comparison across the full distance regime.

Traffic modelling. A model of key request rates, buffer occupancy, and WDM channel sharing would give a more realistic picture of BB84 and MDI at large N, where the direct-mesh assumption becomes economically unphysical.

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